GPG — Practical Exam Notes

UD3: Seguridad lógica — Criptografía y sistemas de identificación

What is GPG? A tool used to encrypt/decrypt files and manage cryptographic keys.

1. Symmetric encryption (password)

Encrypt:

```
gpg -c archivo.txt
```

Uses a passphrase. Creates archivo.txt.gpg.

Decrypt:

```
gpg -d archivo.txt.gpg
```

-c = symmetric encryption; -d = decrypt.

2. Asymmetric encryption (public/private keys)

Public key: share it; other people use it to encrypt files for you. **Private key:** keep it secret; use it to decrypt files sent to you.

Task	Command	Meaning
List keys	`gpg --list-keys`	Shows public keys in your keyring.
Generate key pair	`gpg --full-gen-key`	Creates public and private keys.
Export public key	`gpg --armor --export tu_id_clave > clave.asc`	Saves public key as an armored text file.
Import a key	`gpg --import clave.asc`	Adds key to your keyring.
Edit/sign key	`gpg --edit-key id_clave`	At gpg> type sign then quit.
Encrypt for recipient	`gpg -e -r id_clave archivo.txt`	-e encrypts; -r specifies recipient.
Decrypt recipient file	`gpg -d archivo.txt.gpg`	Recipient needs the matching private key.

3. Fingerprint and trust

After importing a public key, use **gpg --list-keys** to find its fingerprint. Compare it with the owner's fingerprint through a trusted separate channel (e.g. phone or in person). Then sign it:

```
gpg --edit-key id_clave
gpg> sign
gpg> quit
```

To check: reopen with gpg --edit-key id_clave, then type check and quit.

4. Export private key — caution

Command shown in class notes (only for a secure backup):

```
gpg --armor --export-secret-keys tu_id_clave > clave_privada.asc
```

Never share or publish your private key. Store it securely.

5. Keyserver

A keyserver is an internet service that stores and shares public keys so people can find them by email address or key ID. Your class notes mention investigating how to upload a public key to the Ubuntu keyserver.

6. Exam quick cues

Prompt in exercise	Command / response
Encrypt with a password	gpg -c archivo.txt
Decrypt a .gpg file	gpg -d archivo.txt.gpg
List keys	gpg --list-keys
Create public/private keys	gpg --full-gen-key
Send someone your public key	gpg --armor --export tu_id_clave > clave.asc
Import a public key	gpg --import clave.asc
Encrypt for another person	gpg -e -r id_clave archivo.txt

Remember: -c = password encryption; -d = decrypt; --list-keys = list public keys; --full-gen-key = generate keys; -e -r = encrypt for a recipient.

Based on class PDF pages 33–45. Replace example filenames and id_clave with the real names/ID used in the exercise.